

# Week 4 - Quiz

07 May 2025 20:35

The Wiedemann-Franz law is best summarised by which of the following statements?

Select one:

- ☐ a. The toughness and thermal conductivity of materials are linearly related.
- ☐ b. The strength of different materials are linearly related to their density.
- ☐ c. The elastic modulus of different materials are linearly related to their yield strength.
- ☒ d. The electrical and thermal conductivities of materials are linearly related.

Fillers such as talc or calcium carbonate are frequently added to polymer composite materials. Which of the following statements most accurately reflects the motivation for including these types of filler into the compound?

Select one:

- ☐ a. To increase the toughness of the composite material.
- ☒ b. To increase the modulus of the composite material. ✓
- ☒ c. To reduce the density of the composite material. ✗
- ☐ d. To increase the thermal shrinkage from the mould.

Which process involves combining a resin, reinforcement, catalyst and additives in a premixed optimum proportion; the materials are then cut into a shape that is similar to the mould dimensions and the mould is then closed, pressurised and heated to form and cure the part?

Select one:

- ☐ a. Resin transfer moulding
- ☐ b. Hand lay up
- ☒ c. Sheet moulding compound ✓
- ☒ d. Autoclave moulding ✗

Which of the following processes is **not** conventionally used to process thermoplastic polymers?

Select one:

- ☐ a. Blow moulding
- ☒ b. Autoclave moulding ✓
- ☐ c. Extrusion
- ☐ d. Injection moulding

What processing route is usually used to make the plastic cap of a plastic drinking bottle?

Select one:

- ☐ a. Injection blow moulding
- ☒ b. Injection moulding ✓
- ☐ c. Rotational moulding
- ☐ d. Extrusion blow moulding

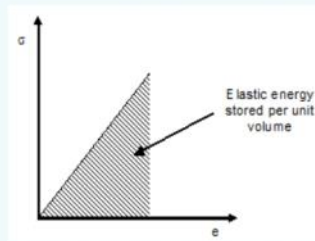
Which of the following does a designer seek to maximise when they design an efficient spring?

Select one:

- ☒ a. Energy stored per unit mass ✓
- ☐ b. The toughness of the spring
- ☐ c. The specific strength
- ☐ d. The modulus

## Materials for springs

Springs come in various shapes, all fulfilling different purposes: there are axial springs (such as a rubber band), leaf springs, helical springs, spiral springs, torsion bars. However, they all possess one functional requirement, which is that of maximising the elastic energy stored per unit volume. The elastic energy stored per unit volume,  $W$ , in a block of material stressed uniformly to a stress  $\sigma$  is the area of the shaded triangle in the figure of stress versus strain of an elastic solid.



Taking the area of a triangle gives the strain energy density for the spring,

$$W = \frac{\epsilon \sigma}{2} \quad (\text{Eq 1})$$

For an elastic solid we also know that

$$E = \frac{\sigma}{\epsilon} \quad (\text{Eq 2})$$

If we take this stress to be the elastic limit  $\sigma_f$  for the materials then it is possible to substitute the strain from equation 2 back into equation 1, which gives

$$W = \frac{\sigma_f^2}{2E}$$

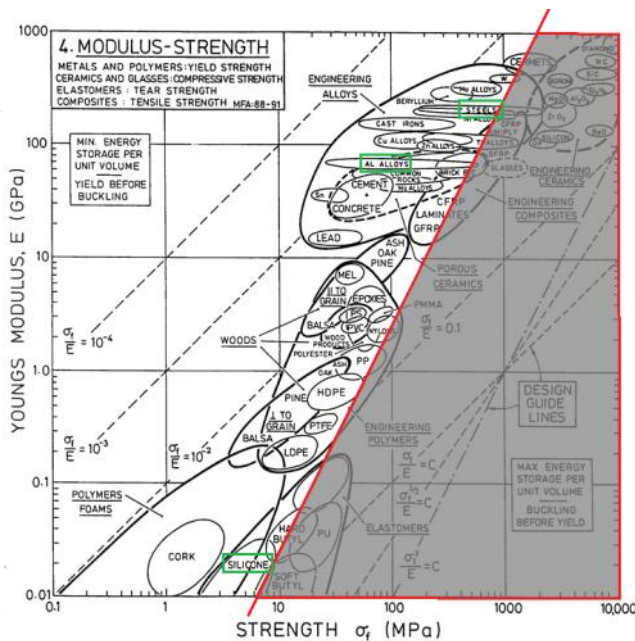
Which of the following is the correct materials index therefore to create a spring that can store the maximum amount of energy in a specific volume.

Select one:

- ☐ a.  $M_1 = \frac{\sigma_f}{E^2}$  ✗
- ☐ b.  $M_1 = \frac{\sigma_f}{E^3}$  ✗
- ☐ c.  $M_1 = \frac{\sigma_f}{E}$  ✗
- ☒ d.  $M_1 = \frac{\sigma_f^2}{E}$  ✓

$W \uparrow$  maximise

$\therefore \frac{\sigma_f^2}{E} \uparrow$  maximise



Steels  
Al-Alloys  
Silicone

#### Materials for springs (minimum mass)

In practice sometimes we are not interested in the energy stored per unit volume but the energy stored per unit mass of spring. In this case we take the previous equation for the minimum volume spring and divide it by the density of the material.

Which of the following is the correct materials index therefore to create a spring that can store the maximum amount of energy per unit mass.

Select one:

☐ a.  $M_2 = \frac{\sigma_f}{\rho E^3}$  ✗

☐ b.  $M_2 = \frac{\sigma_f}{\rho E}$  ✗

☐ c.  $M_2 = \frac{\sigma_f}{\rho E^2}$  ✗

☐ d.  $M_2 = \frac{\sigma_f^2}{\rho E}$  ✓

$$M = \frac{\sigma_f^2}{\rho E}$$